

Alvaro Alejandro Cruz Perez

(928) 941-8709 | aacp1519@gmail.com | Tucson, AZ | www.linkedin.com/in/alvaro-cruz-perez-540413274

EDUCATION

Accelerated Master of Science in Biomedical Engineering | Expected May 2027

Bachelor of Science in Biomedical Engineering | May 2026

The University of Arizona, College of Engineering | Tucson, Arizona

GPA: 3.43

Relevant Coursework: Medical Device Design; Biological and Synthetic Materials; Biomaterial-Tissue Interactions; Biomedical Imaging; Biomedical Statistics

Research Programs and Honors: Undergraduate Biology Research Program (UBRP) Scholar; Place-based Health Informatics Research Experience (PHIRE) | Dean's List Spring/Fall 2024, Fall 2025, Spring 2026

PROFESSIONAL EXPERIENCE

Dr. Hutchinson's Multi-scale Brain Imaging Lab, University of Arizona | Tucson, Arizona

Ferret Research Team Lead | August 2026 - Present

- Lead day-to-day coordination of the ferret research team, assigning shared responsibilities, organizing routine animal care and maintenance, and supporting ongoing research activities.
- Train and mentor team members on research procedures, project workflows, and troubleshooting while supporting consistency across shared laboratory responsibilities.

Research Assistant | May 2024 – Present

Ferret Behavior Project – Automation, Kinematics, Computer Vision & Data Analysis

- Develop a markerless gait-analysis workflow using **DeepLabCut** and Python to quantify locomotion from multi-camera video, including walking speed, step/stride characteristics, joint angles, and angular velocity.
- Refine video acquisition, model training, filtering, and validation procedures to improve pose-estimation reliability across subjects, recording conditions, and longitudinal timepoints.
- Conduct comprehensive literature reviews to identify research gaps and inform future experimental design.

Ferret Cradle Design Project – Research Equipment Development, Mechanical Design, Prototyping & Validation

- Engineered a **3D-printed-MRI-compatible** ferret positioning cradle by translating user needs from veterinary staff and MRI technicians into functional specifications in **SolidWorks**.
- Performed iterative prototyping and **validation** on a 7T MRI system, ensuring compatibility, reliability, and repeatable performance.
- Designed a modular rail-and-hook attachment system that improved upon commercial solutions, increasing usability and adaptability for multi-modal imaging.
- Collaborated with cross-functional teams (engineering, veterinary, imaging staff) to refine device design and meet operational constraints.

RELEVANT ENGINEERING PROJECTS

Mechanical Engineering Lead-MACRO: Multidigit Assessment of Choice Response Output | Fall 2025 - Spring 2026

ENGR498A/B - Interdisciplinary Capstone | University of Arizona

- Supported design and development of an electromechanical hand assessment device integrating sensors (FSRs) and actuators (LRAs), guaranteeing a standardized and reliable procedure.
- Led **mechanical design** of an ergonomic, adjustable digit-size hand block for multidigit choice response assessment system, accommodating 95% of the population.
- Created SolidWorks **engineering drawings** with dimensions and tolerances for custom components and assemblies to support fabrication, assembly, and design communication.

Medical Device Design – Heat Illness Detection Wearable | Spring 2025

BME 310: Medical Device Design | University of Arizona

- Integrated pulse-oximetry, accelerometer, and temperature sensors and used systematic root-cause analysis to diagnose communication errors, verify voltages, isolate faulty components, and complete circuit-level rework.

CERTIFICATIONS

CITI Program: Biomedical Research Investigators (expires Feb 2029) | Responsible Conduct of Research-Basic (expires May 2027) | Working with the IACUC-Investigators, Staff and Students

University of Arizona: Science Communication-Undergraduate Biology Research | Research Readiness

TECHNICAL SKILLS

Research & Development: SolidWorks, 3D printing, Structural Finite Element Analysis fundamentals (coursework), soldering, Eagle PCB, hardware integration, engineering drawings, dimensioning/tolerancing, experimental data collection, literature review, technical documentation, and troubleshooting

Data Analysis and Programming: Python, MATLAB, R, C++, Excel, DeepLabCut

Collaboration and Communication: Cross-functional research teams, scientific presentations, technical reports, fluent Spanish